EE426 / 626 Microprocessor Applications

Some History



EE426 / 626 Microprocessor Applications

Microprocessor (MPU or μP)



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Microprocessor (MPU or μP)

A microprocessor is a complete central processing unit — arithmetic/logic unit (ALU), control unit, and instruction decode/registers — implemented on a single integrated circuit (one die), capable of executing a program of instructions fetched from memory.

— Single-chip distinguishes it from a CPU built from multiple ICs (a "chipset"/"bit-slice" CPU).

Purists exclude MP944 from the strict definition even though it predates the i4004

— General-purpose / programmable distinguishes it from fixed, hard-wired logic built for one task.

A distinction Intel's Ted Hoff drew against Busicom's original 7-chip hard-wired design.

Under this strict single-die reading, the 4004 has the cleaner claim; MP-944 and TI's TMX 1795 each fail on a different technicality (multi-chip, and never-shipped / prototype respectively)

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Microcontroller (MCU or μC)



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Microcontroller (MCU or μC)

A microcontroller is a single IC that packages a microprocessor core together with the memory and peripherals needed to run a standalone embedded application.

— Typically has on-chip RAM, on-chip program memory (ROM/Flash/EPROM)

— On-chip peripherals such as I/O (timers, serial ports, ADCs, GPIO)

TI's TMS 1000 (1974) is usually credited as the first commercial single-chip microcontroller rather than microprocessor, for exactly this reason — it bundled ROM, RAM, and I/O on-die for calculator control, whereas the 4004 needed the surrounding 4001/4002/4003 chipset to form a complete system.



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Early Timeline

Robert Noyce, *Gordon Moore* — Cofounded Fairchild in 1957

Federico Faggin — Developed: Self aligned silicon gate and buried contacts (1968)

Noyce and Moore — Leave Fairchild to found Intel (1968)

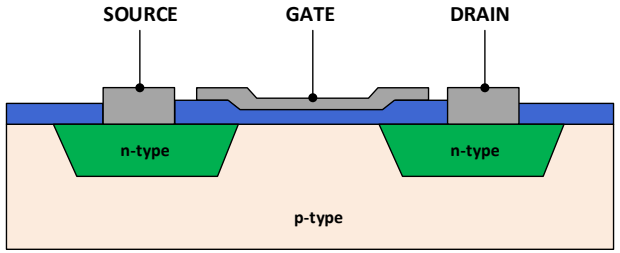
Federico Faggin — Leaves Fairchild to join Intel (1970)

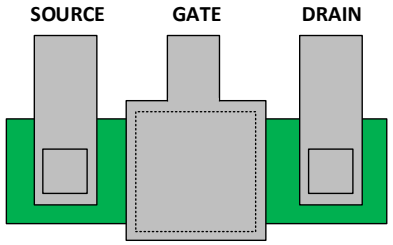
Masatoshi Shima — Leaves Busicom to join Intel (1970)

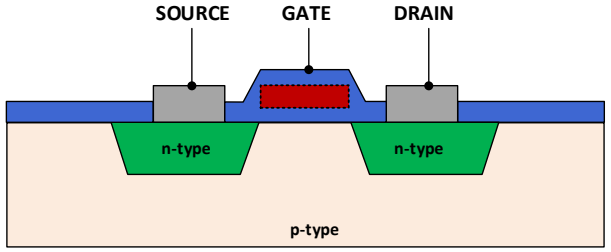
Develop MCS-4 Chipset

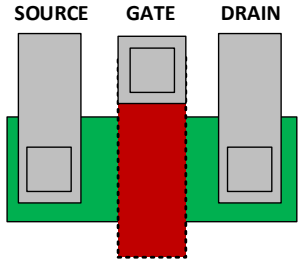
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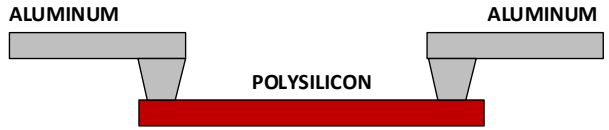
(Poly) Silicon Gate Technology (SGT) and Buried Contacts

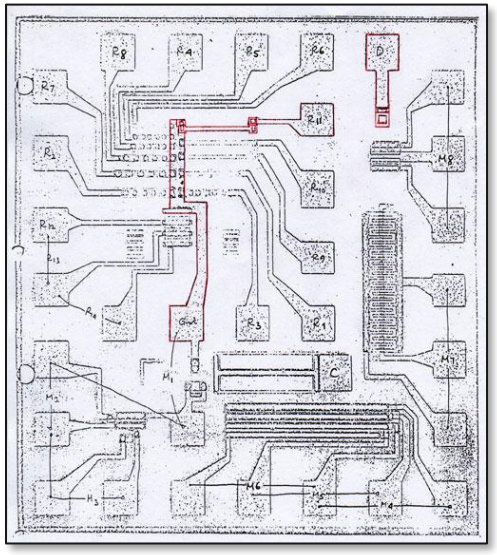








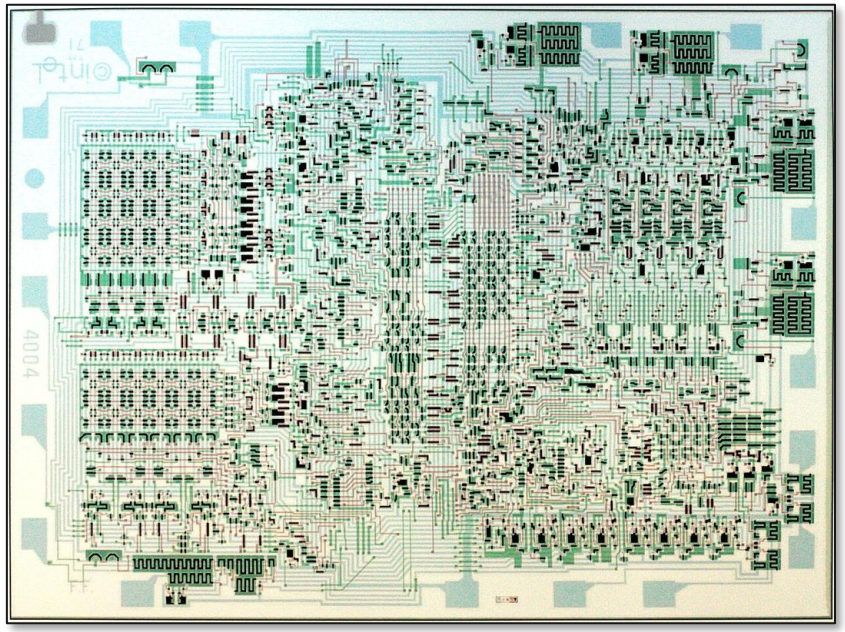




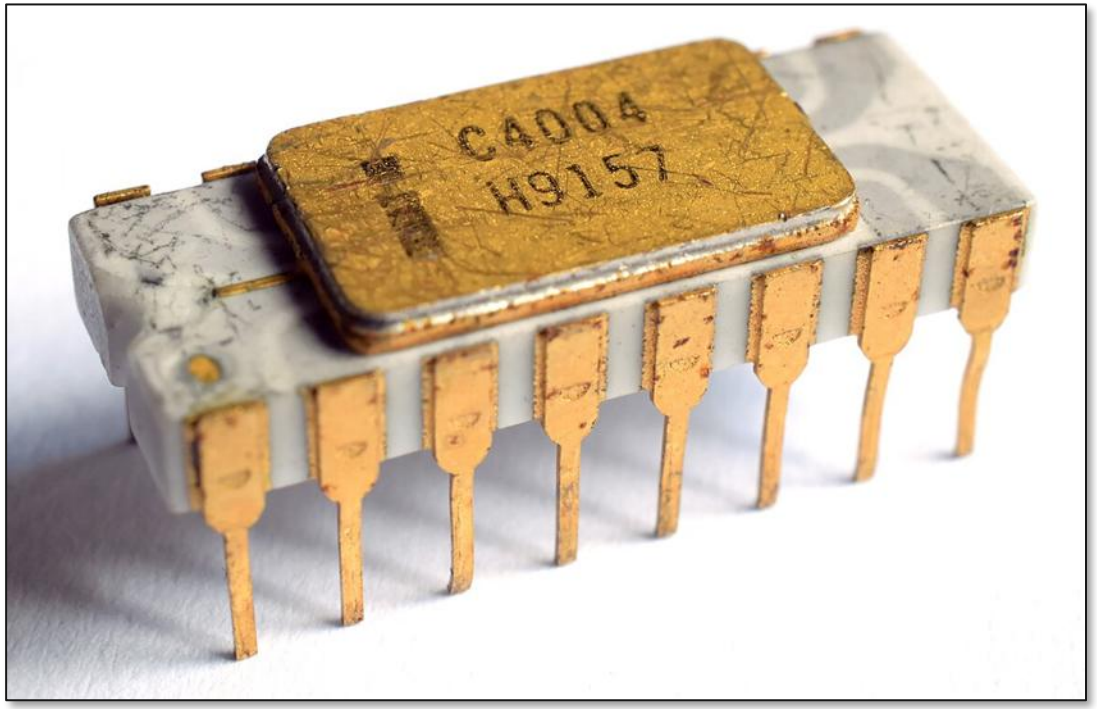
http://www.intel4004.com/buried.htm

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Intel 4004



By Intel 4004 Chip Layout (29983532570).jpg: Wolfgang Stief from Tittmoning, Germanyderivative work, the rectangular version above: Pittigrilli - Intel 4004 Chip Layout, CC0, https://commons.wikimedia.org/w/index.php?curid=147243300



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Busicom Corp

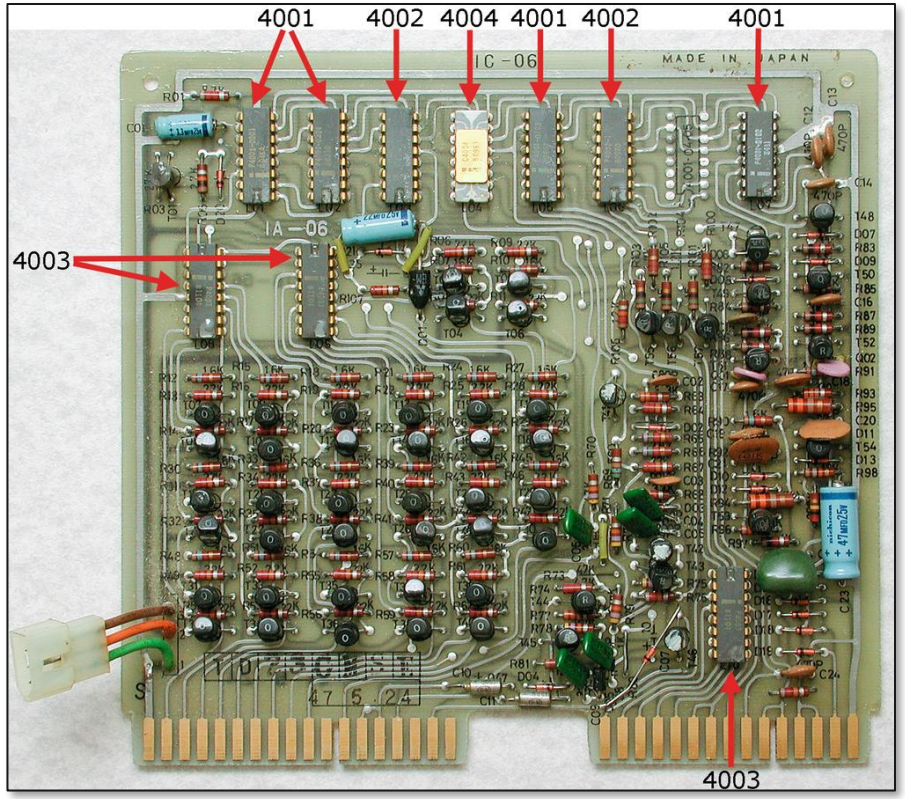
Intel MCS-4 (4001, 4002, 4003, 4004)

Intel 4001	256-byte 4-bit ROM
Intel 4002	40 byte DRAM
Intel 4003	I/O chip with 10-bit shift register
Intel 4004	4-bit CPU

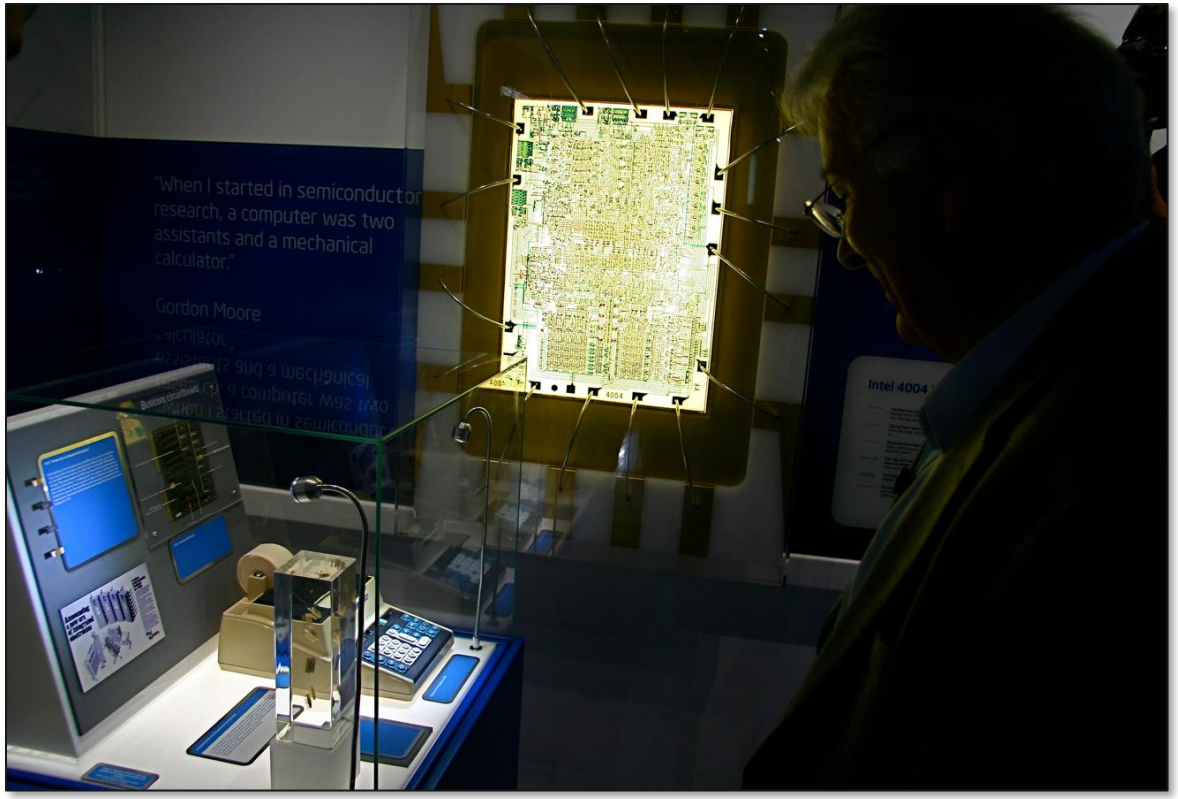


By Christian Bassow, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=55532794

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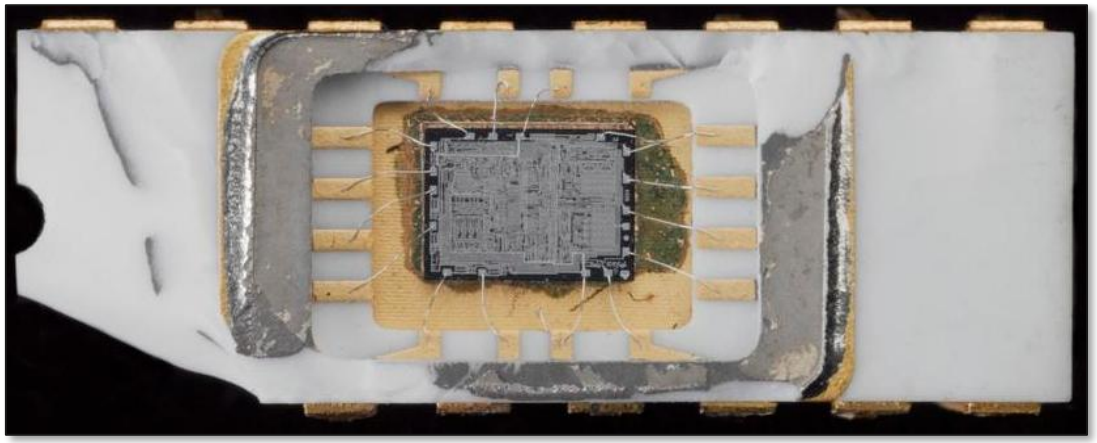
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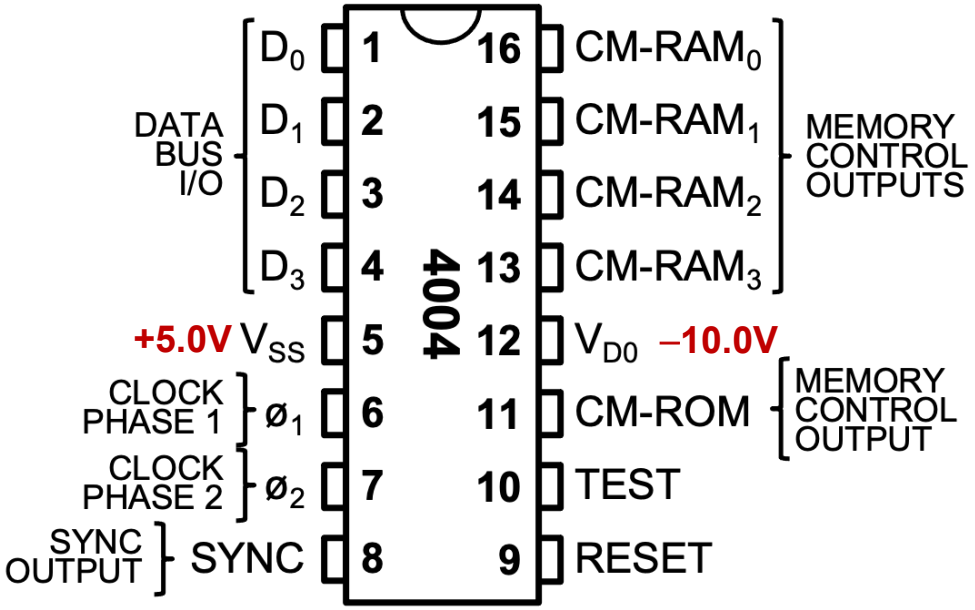
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Intel 4004 – 740 kHz Clock Speed



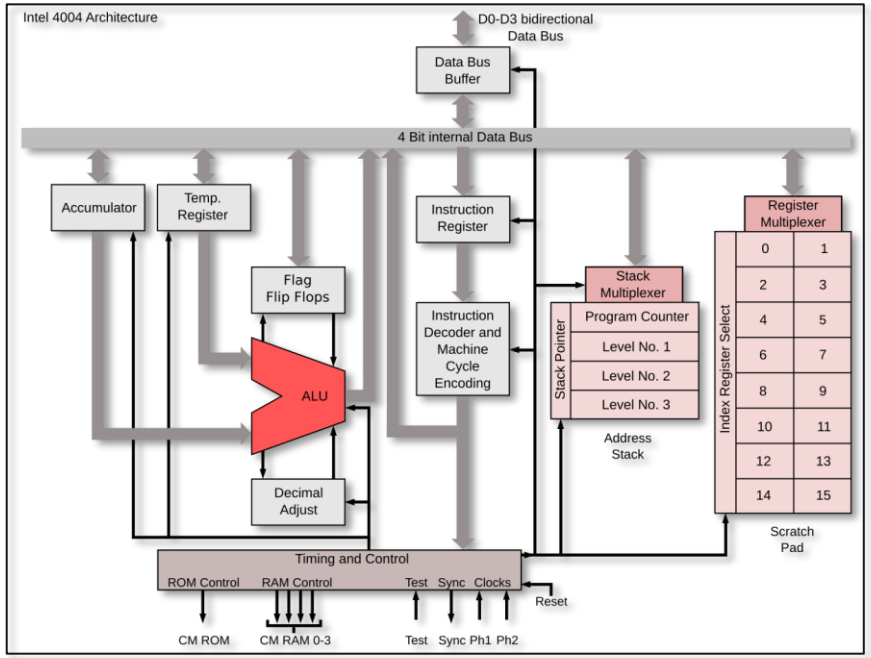
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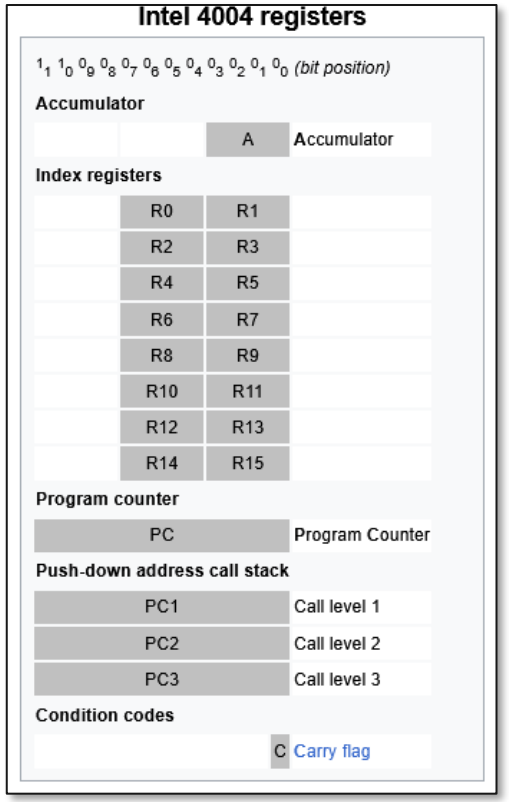
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Intel 4004



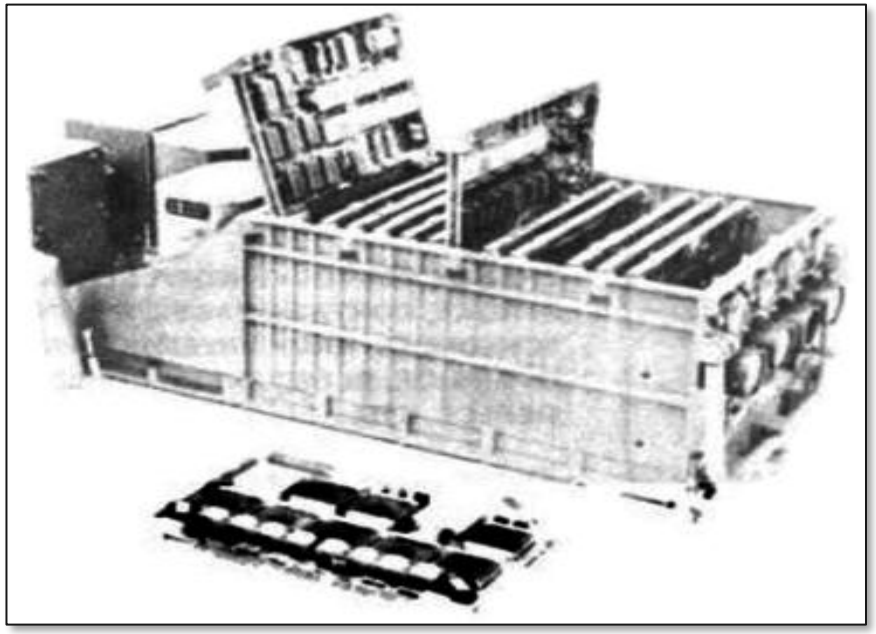
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From Wikipedia, the free encyclopedia
https://en.wikipedia.org/wiki/Intel_4004

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MP944 Garret AiResearch



By Department of the Navy - Fiscal Year 1973 Authorization for Military Procurement, Pt. 6, Page 4130, Public Domain, https://commons.wikimedia.org/w/index.php?curid=67584674



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TMS 1000 Texas Instruments (1974)



By FozzTexx at English Wikipedia, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=79574006



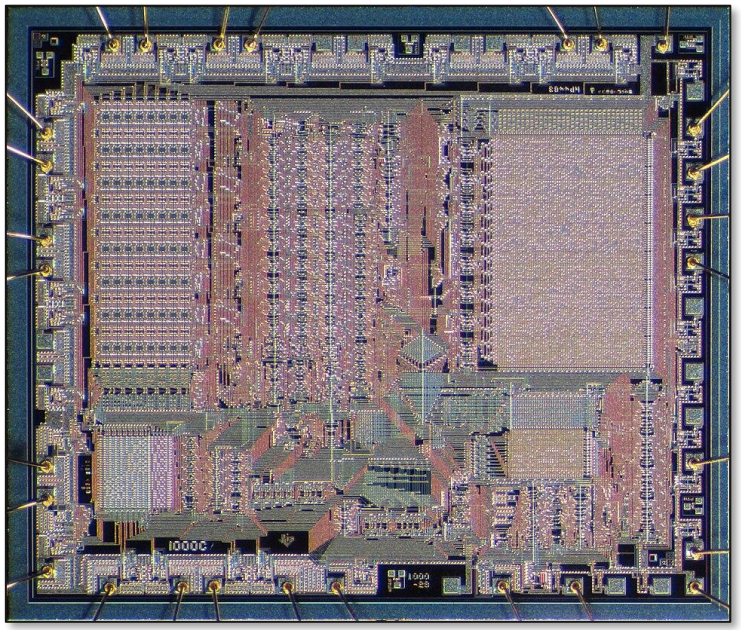
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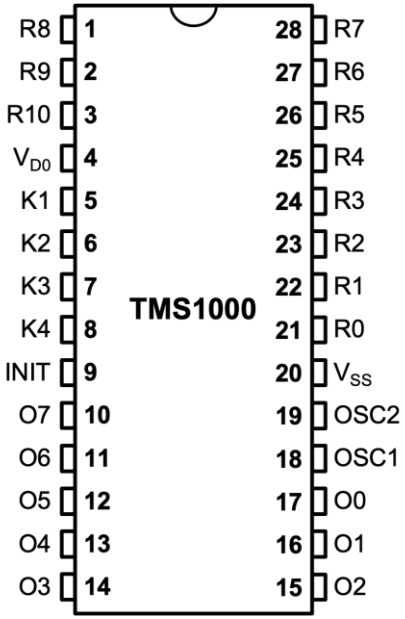
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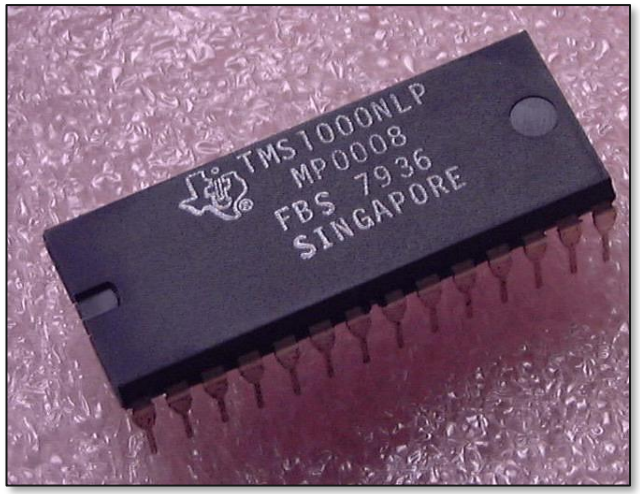
TMS 1000 Texas Instruments (1974)



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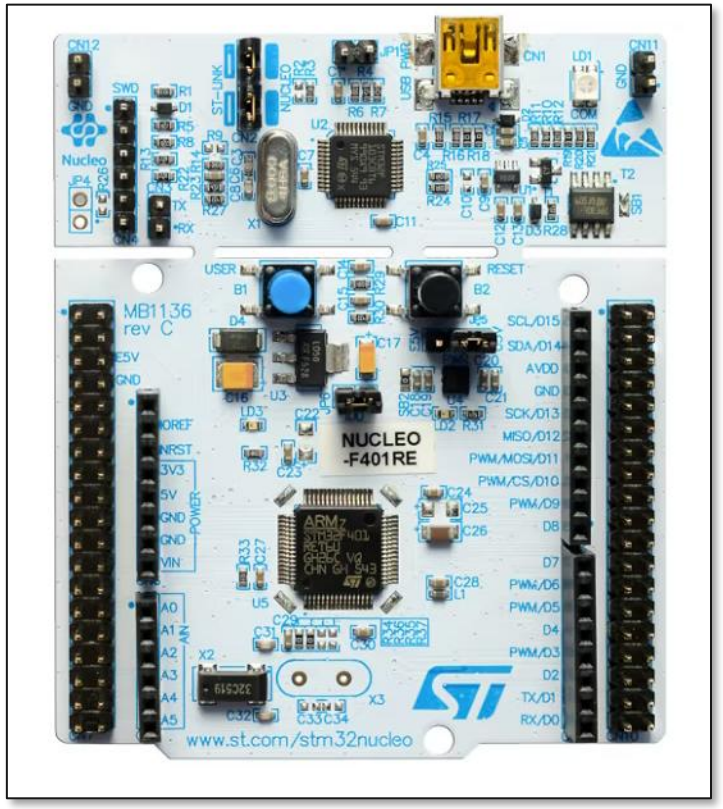
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Fast Forward ...



https://www.st.com/en/evaluation-tools/nucleo-f446re.html

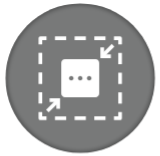
ARM Cortex-M4 core with FPU, up to 180 MHz
Adaptive real-time (ART) memory accelerator with prefetch
Up to 512KB Flash memory
128KB SRAM plus 4KB backup SRAM
Dual-mode Quad-SPI memory interface
Crypto/hash processor: AES 256-bit, HASH, true RNG
Digital camera interface (DCMI)
SDIO interface for SD/MMC/eMMC
Up to 4 USARTs and 4 UARTs
Up to 4 SPI interfaces, 2 with I2S support
2 Serial Audio Interfaces (SAI)
I2C1, I2C2, I2C3 (standard/fast mode) plus FMPI2C1 (Fast Mode Plus, 1 MHz)
2 CAN 2.0B interfaces
USB OTG full-speed and high-speed controllers
Up to 17 timers, including 2 advanced-control and 12 general-purpose
3 independent 12-bit ADCs, up to 24 channels
2 12-bit DACs
Independent and window watchdog timers
RTC with calendar and hardware calibration
CRC calculation unit
96-bit unique device ID
Up to 114 fast GPIO pins
Operating voltage range 1.7V to 3.6V
Multiple package options (LQFP, BGA, WLCSP)

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Segger J-Link EDU Mini



https://www.segger.com/products/debug-probes/j-link



Small size

With dimensions of just 18 mm × 50 mm, the J-Link EDU Mini is similar in size to a USB stick and fits easily into any workspace or laptop bag.



Fast downloads

Download speeds of up to 200 KB/s help reduce programming times during development and testing.



Full J-Link functionality

The J-Link EDU Mini provides essential J-Link debugging and programming capabilities in a compact form factor for educational use.



Broad MCU support

Supports a broad range of microcontrollers and provides unlimited flash breakpoints for debugging directly in flash memory.

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Segger Ozone



https://www.segger.com/products/development-tools/ozone-j-link-debugger

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Team Roles

Program Manager

- Owns overall schedule — tracks milestones: Proposal, PDR, CDR, Final
- Owns scope and fallback scope — decides what gets cut if behind schedule
- Owns budget / BOM tracking — sourcing, cost, availability across dual-source parts
- Primary point of contact with instructor / reviewers
- Runs team meetings, assigns action items, tracks closure at PDR / CDR
- Manages risk — flags blockers early, reallocates work as needed
- Owns final marketing brochure deliverable
- Accountable for on-time, on-budget completion of the project



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Team Roles

Technical Lead / Hardware Developer

- Owns overall technical architecture and design decisions
- Owns hardware design — schematic capture, PCB layout, component selection
- Owns peripheral checklist and hardware bring-up
- Resolves technical tradeoffs between team members' subsystems
- Leads PDR / CDR technical content and action-item closure
- Interfaces between hardware and firmware / software developers
- Owns final technical specification deliverable

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Team Roles

Firmware Developer

- Owns low-level firmware — peripheral drivers, register-level configuration
- Owns interrupt and DMA-driven I/O implementation
- Owns toolchain and version-control discipline for firmware codebase
- Integrates firmware with hardware bring-up alongside Tech Lead
- Leads PDR / CDR firmware content and action-item closure
- Owns watchdog timer + low-power sleep mode implementation (shared with Tech Lead)



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Team Roles

Application / Firmware Developer

- Owns application-level software — user-facing logic, data processing, UI / UX if applicable
- Owns integration between application layer and firmware layer
- Owns communication protocols (UART/ USB) at the software-application level
- Supports firmware bring-up alongside Software Dev 2
- Leads PDR / CDR application-software content and action-item closure
- Interfaces with Tech Lead on hardware-software boundary decisions
- Owns final user's manual deliverable

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Project Ideas

1. Mini oscilloscope/logic analyzer — ADC+DMA continuous capture, UART stream to a laptop viewer, interrupt-triggered capture.
2. Reaction-time tester — interrupt-timed button press, PWM/LED cue, timer-measured latency, UART/display scoreboard.
3. Environmental data logger — I2C/SPI sensor, timer-interrupt sampling, DMA buffering, sleep between samples.
4. DAC waveform synth — ADC-read potentiometer selects waveform/frequency, DMA-driven continuous DAC output.
5. Morse encoder/decoder — interrupt-timed button input, UART/buzzer decode; bonus two-board transceiver pair.
6. Digital lock/access control — GPIO keypad or encoder, state machine, I2C/SPI display, sleep between interactions.
7. Closed-loop motor/fan speed controller — PWM drive, timer input-capture or ADC feedback, UART telemetry.
8. Two-board sensor node + base station — sleep/wake sampling over I2C, UART/SPI link to a logging board.
9. Audio-ish VU meter — DMA-sampled ADC, envelope calc, PWM/LED-bar display; FPU/DSP as optional stretch.
10. Six-axis accelerometer/IMU device — motion sensing via I2C or SPI; bubble level, step counter, or tap/gesture detection.

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Project Ideas

11. Small game console — GPIO buttons, SPI/I2C display or LED matrix, timer game loop, PWM buzzer.
12. Host-dashboard telemetry link — UART stream to a Python dashboard, optional bidirectional actuator control.
13. GPS location/tracking display — parse NMEA sentences, show/log position, speed, time.
14. WiFi survey tool — scans networks via SPI-based WiFi module, displays SSID/channel/RSSI/security, mirrors to UART.
15. Cellular network survey tool — queries UART/AT-command cellular module for signal strength, displays on OLED and UART.
16. Servo-controlled pointer/pan-tilt mount — PWM-driven RC servo via ADC potentiometer/joystick or UART commands.
17. Solar-powered remote temperature & humidity datalogger — RTC wakeup duty-cycling, I2C sensor, solar+battery powered.
18. Real-time signal effects processor — continuous ADC / DAC pipeline via DMA, real-time filters and delays.
19. RGB/addressable lighting controller — PWM/bit-banged WS2812-style output, ambient auto-brightness.
20. Ultrasonic distance/proximity instrument — GPIO trigger, timer input-capture echo measurement, buzzer/servo feedback.



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Project Ideas

21. Serial command-line instrument controller — UART terminal controls/queries an I2C or SPI peripheral.
22. Audio alarm clock — RTC scheduling, WAV/PCM playback via SD/SPI + DMA to DAC/SAI, GPIO buttons.
23. NFC/RFID access logger — SPI/I2C NFC reader, RTC-timestamped access events, logs to storage or OLED.
24. Secure element / crypto co-processor — SPI/I2C peripheral for keypair gen, sign, verify, encrypt/decrypt.
25. GPS-disciplined clock — GPS 1PPS via GPIO interrupt disciplines RTC against crystal drift.
26. DTMF encoder/decoder — DAC-generated touch-tones, ADC-sampled Goertzel decoding.
27. FSK modulator / demodulator — UART data to FSK tone stream via DAC / DMA, ADC demodulation back to UART.
28. Sub-GHz data sniffer (e.g. 315 MHz TPMS) — SPI sub-GHz receiver, decodes pulse timing, UART reports payload
29. Pulse oximeter — I2C SpO2/heart-rate sensor, interrupt-driven sampling, display or UART output. Educational prototype only.
30. Blood glucose meter — ADC reads test-strip signal, calibration lookup, display, UART logging. Educational prototype only.

